Using DIY sensors to monitor air quality in urban areas

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Introduction

Did you know that in many cities around the world have a lack of knowledge of their air quality due to the low importance they give to this and the high cost of the monitoring station?

Air pollution remains one of the greatest environmental health risks globally. Pollutants of major public health concern include particulate matter, carbon monoxide, ozone, nitrogen dioxide and sulfur dioxide. Outdoor and indoor air pollution cause respiratory and other diseases and are important sources of morbidity and mortality. An increasing number of low-cost and do-it-yourself (DIY) digital sensors for monitoring air quality are now in circulation (Pritchard, Gabrys, & Houston, 2018). The devices are easy to deploy and can complement data from official networks, which rely on sophisticated but sparsely distributed sensors. The ‘citizen-science’ approach aims to provide high-resolution measurements of air pollution where people actually live (Austen, 2015).

In areas such as Río Abajo and Cerro Viento, within the Panama City metropolitan region. There is limited information that talk about the spatial temporal patterns of air pollution in the area. As an example, Rio Abajo the air quality is generally acceptable for the majority of individuals. But this doesn’t talk about the microenvironments of the area, meanwhile Cerro Viento lacks a systematic resolution information on public reports.

Some challenges this presents is the high cost of the majority of devices made for monitoring air pollution and the lack of data provided according to this problem. (DIY) efforts are part of a push to democratize air-quality monitoring so that it no longer remains solely in the domain of governments and academic researchers (Austen, 2015).

How can the data shown by the sensor stablish a difference between air quality in Cerro Viento and Rio Abajo, and how does this difference have to do with some factors such as traffic density?

Objectives

Main

To analyze differences in air quality between Cerro Viento and Río Abajo using data from a DIY sensor, and relate these differences to factors such as traffic density.

Specific

1. To measure and record PM2.5 concentrations at selected locations in Cerro Viento and Río Abajo using the DIY sensor.

2. To compare the air quality data from both sites and identify observed differences.

3. To evaluate how traffic density and other local factors may contribute to the variations in air pollution between the two areas.

Methodology

This project used an experimental design to collect and compare air quality data in Cerro Viento and Río Abajo, using a DIY air quality sensor. The approach involved measuring PM2.5 concentrations at selected sites and analyzing how traffic and local environmental factors affects spatial differences in pollution levels.

Materials:

- SDS011 PM Sensor
- Laptop
- Visual Studio Code
- Power BI

Variables:

Independent Variable:

- Location of measurement (Cerro Viento vs. Río Abajo)

Dependent Variable:

- PM2.5 concentration (µg/m^3)

Controlled Variables:

- Type of sensor used
- Measurement method
- Sampling duration and frequency
- Time of day of measurements
- Weather conditions
- Sensor placement height

Setup & Calibration:

Not necessary

Procedure:

1. We developed a python script for reading the sensor's data and its storage in text files.
2. The SDS011 sensor was connected to the computer and runned the python script for 1 hour
3. The data was processed in power BI to create the graphs.

Data:

In September 28, 2907 (PM2.5) measurements were obtained, each data was recorded every second, in Cerro Viento, from 3:12pm to 4:01pm. (48 minutes)

In September 29, 1561 (PM2.5) measurements were obtained, in Rio Abajo, each data was recorded every second also, from 9:19 to 9:59. (30 minutes)

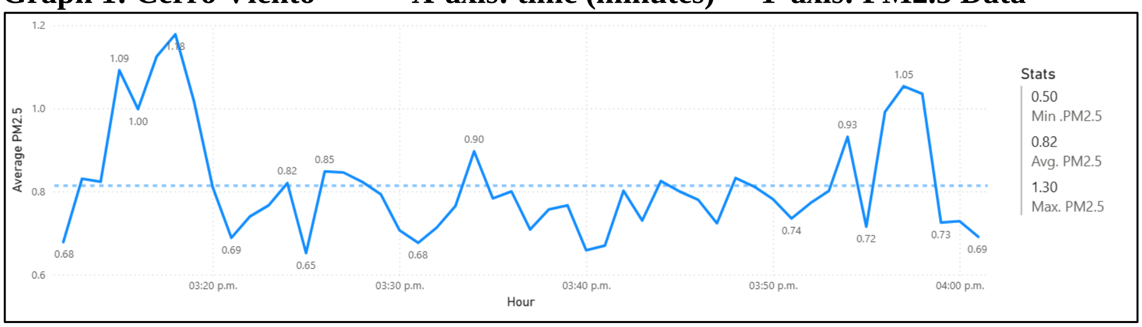
Quality Assurance / Changes:

We eliminated the data PM10 because it was the same measurements in both sides, therefore, these measurements were excluded as they were errors in the sensor reading or program operation.

In the graphs the sensor for any reason stops getting measurements, for any reason between 10pm up to 10:20 pm, so we decided to use the data from Cerro Viento's graph and adjust it until it gets to the same amount of time as the Rio Abajo graph.

Results

Graph 1: Cerro Viento X-axis: time (minutes) Y-axis: PM2.5 Data



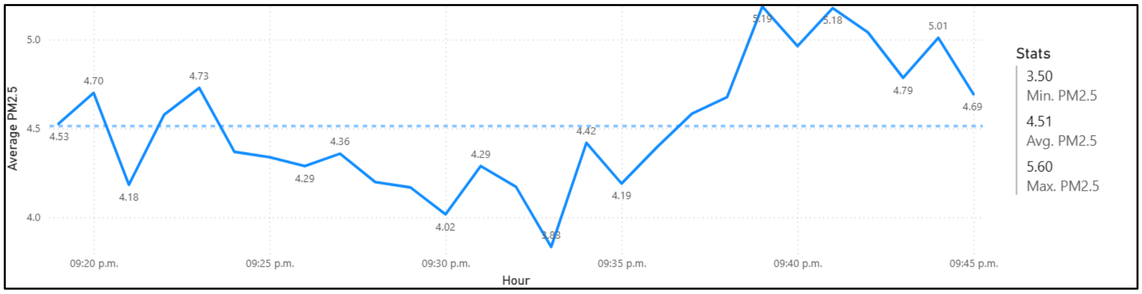
Minimum PM2.5: 0.5

Maximum PM2.5: 1.3

Average PM2.5: 0.82

The graph shows that PM2.5 levels in Cerro Viento remained relatively low and stable, with an average of 0.82 µg/m³ and small fluctuations throughout the sampling period.

Graph 2: Rio Abajo   X-axis: time (minutes)   Y-axis: PM2.5 Data



Minimum PM2.5: 3.5

Maximum PM2.5: 5.6

Average PM2.5: 4.51

The graph shows that PM2.5 levels in Rio Abajo were consistently higher, averaging 4.51 µg/m³, indicating poorer air quality compared to Cerro Viento.

Discusssion

The results show that Río Abajo had significantly higher PM2.5 concentrations (average = 4.51 µg/m³) compared to Cerro Viento (average = 0.82 µg/m³), indicating poorer air quality in the Río Abajo area. This suggests that local environmental factors, especially traffic density, strongly influence air pollution levels between the two sites.

These findings align with previous research by Austen (2015) and Pritchard, Gabrys, and Houston (2018), who reported that urban areas with higher vehicle activity and limited air circulation tend to exhibit elevated particulate matter concentrations. Similar studies have also shown that traffic emissions are a dominant source of PM2.5 in densely populated regions.

The observed difference between the two sites may result from heavier traffic flow and reduced vegetation coverage in Río Abajo, which contribute to the accumulation of airborne particles. In contrast, Cerro Viento, being more residential and less congested, likely benefits from greater air dispersion and lower emission sources.

Some limitations of this study include the short sampling period and the limited number of sensors, which might not capture long-term variations or microenvironmental differences.

Additionally, meteorological factors such as wind direction and humidity could have influenced the readings.

Overall, these results demonstrate the usefulness of accessible, low-cost air sensors for gathering neighborhood-level air quality data. Understanding these spatial variations can help communities and policymakers develop targeted strategies to reduce pollution exposure and promote healthier urban environments.

Conclusion

The objectives were achieved. PM2.5 levels were successfully measured in Cerro Viento and Río Abajo using the DIY sensor. The results showed that Río Abajo had consistently higher PM2.5 concentrations than Cerro Viento. These differences align with observed variations in traffic density and local activity, confirming that higher traffic levels contribute to poorer air quality in Río Abajo.

Reflection

For our experiment, we used habit 2: Begin with the End in Mind, because from the start, we clearly had our goal. Having that, it helped us plan and organize our project. We also used habit 6: Synergize, each member had a role, supporting each other to make sure we were on the right path for our project. And lastly, we also used habit 1: Be Proactive, because we often faced difficulties but had to find a way to make ourselves work on a solution, to reach the goal we had from the start.

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